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Review

Evolutionary techniques in optimizing machining parameters: Review and recent applications (2007–2011)

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ABSTRACT

In highly competitive manufacturing industries nowadays, the manufactures ultimate goals are to produce high quality product with less cost and time constraints. To achieve these goals, one of the considerations is by optimizing the machining process parameters such as the cutting speed, depth of cut, radial rake angle. Recently, alternative to conventional techniques, evolutionary optimization techniques are the new trend for optimization of the machining process parameters. This paper gives an overview and the comparison of the latest five year researches from 2007 to 2011 that used evolutionary optimization techniques to optimize machining process parameter of both traditional and modern machining. Five techniques are considered, namely genetic algorithm (GA), simulated annealing (SA), particle swarm optimization (PSO), ant colony optimization (ACO) and artificial bee colony (ABC) algorithm. Literature found that GA was widely applied by researchers to optimize the machining process parameters. Multi-pass turning was the largest machining operation that deals with GA optimization. In terms of machining performance, surface roughness was mostly studied with GA, SA, PSO, ACO and ABC evolutionary techniques.

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1. Introduction

In manufacturing, the process of removing unwanted segment of metal workpiece in the form of chips is known as machining. Machining is one of the five groups of manufacturing processes which includes casting, forming, powder metallurgy and joining (Nagendra Parashar & Mittal, 2007). The machining process will shape the workpiece as desired and it is usually done using machine and cutting tools. The machining cutting process can be divided into two major groups which are (i) cutting process with traditional machining (e.g., turning, milling, boring and grinding) and (ii) cutting process with modern machining (e.g., electrical discharge machining (EDM) and abrasive waterjet (AWJ)). From the early introduction cannon-borring machine by John Wilkinson in 1775 to a modern machine CNC (Computer Numeric Control) in the 1960s, the machining processes continues to evolve where new techniques and modern tools have been discovered. There are many researches that have been done in the areas of machining processes which mainly stressed on the tool, input work materials and machine parameter setting (Mukherjee & Ray, 2006). In the paper, a review on the optimization techniques in metal cutting processes are focusing on (i) modeling techniques and (ii) conventional and non-conventional (evolutionary) optimization techniques as illustrated in Fig. 1. This study also pointed out that modelling and optimization techniques have been applied in the recent research due to the complexity of mathematical model to determine optimal machining process parameters. It was reported that evolutionary techniques such as GA, SA and ACO for optimization process parameters have been applied in the traditional machining due to likely to deal with highly nonlinear, multidimensional and ill-behaved complex engineering problem (Chandrasekaran, Muralidhar, Krishna, & Dixit, 2010; Mukherjee & Ray, 2006). In the review paper by Benardos and Vosnaikos (2003), the authors provided an evaluation based on machining theory, experimental investigation, design of experiments (DOE) and artificial intelligence (AI) techniques in optimizing machining process parameters. In the literature review paper by Aggarwal and Singh (2005), the authors discussed the various conventional techniques (e.g., geometric programming and goal programming) and evolutionary techniques (e.g., GA) in optimizing traditional machining process parameters, turning operation.

In optimizing the machining process parameters, the selection of machining process parameters is a very crucial part in order for the machine operations to be successful (Rao & Pawar, 2010b). To choose the process parameters, it is usually based on the human (or manufacturing engineers) judgement and experience. However, the chosen process parameters usually did not give

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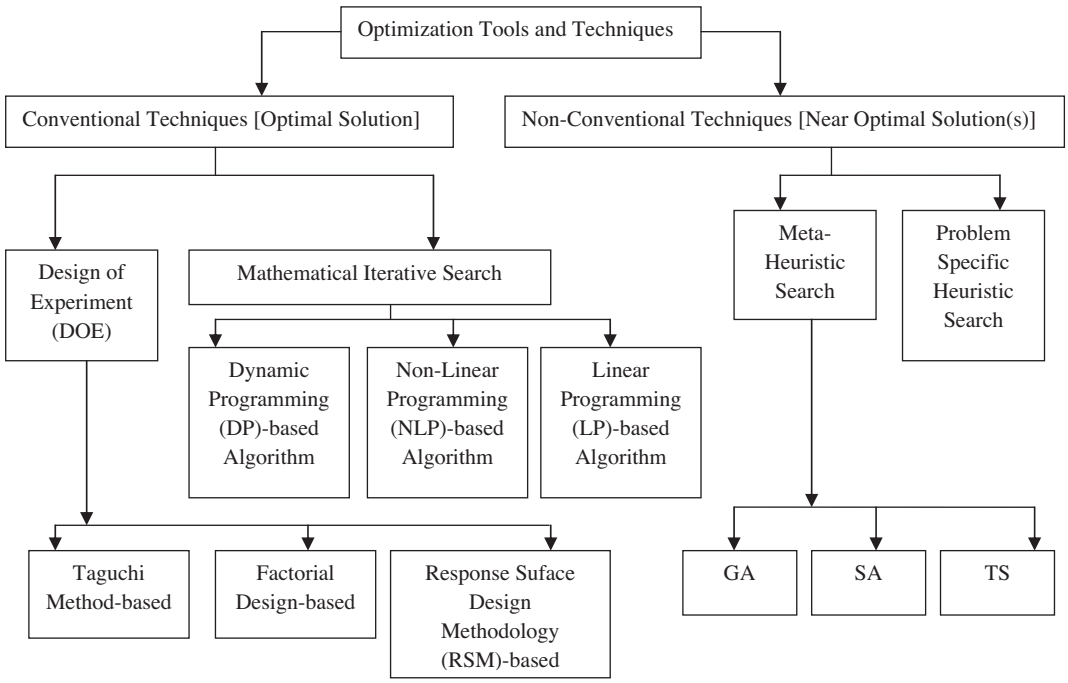

Fig. 1. Conventional and non-conventional optimization tools and techniques (Mukherjee & Ray, 2006).

an optimal result. This is due to the fact that in machining processing; a number of factors also could interrupt thus preventing in achieving high process performance and quality (Benardos & Vosnaikos, 2003). In fact, tuning each machining process parameters would give significant effects to others parameters as well.

In the current trends of optimizing machining process parameters, various evolutionary or meta-heuristic techniques have been used. Most of these techniques are inspired by nature or animal behaviour such as GA, PSO ACO and ABC. According to Vob (2001), the definition of meta-heuristic technique is an iterative master process that guides and modifies the operation of subordinate heuristics to efficiently produce high-quality solutions. It may manipulate a complete (or incomplete) single solution or a collection of solutions at each iteration. The subordinate heuristics may be high (or low) level procedures, or a simple local search, or just a construction method. The family of meta-heuristics includes, but is not limited to, adaptive memory procedures, tabu search (TS), ant systems, greedy randomized adaptive search, variable neighborhood search, evolutionary methods, GA, scatter search, neural networks, SA, and their hybrids. The most recent research of evolutionary techniques in machining process parameters optimization have been demonstrated by Rai, Brand, Slama, and Xirouchakis (2011), Gao, Li, and Mao (2011), Rao and Pawar (2010b), Sultana and Dhar (2010), Wang, Yuan, Hu, and Dengn (2009) and Zhang and Chen (2009). In this paper, we discuss five evolutionary techniques (GA, SA, PSO, ACO and PSO) and basic methodology of each technique in optimizing machining process parameters for both traditional and modern machining.

2. Genetic algorithm

According to Ganesan, Mohankumar, Ganesan, and Ramesh Kumar (2011), GA and PSO is one the best population search techniques. GA optimization technique has been used by a number of researchers to find the optimal surface roughness in various traditional and modern machining (Maji & Pratihar, 2010; Pasam, Battula, Valli, & Swapna, 2010; Wang et al., 2009; Zain, Haron, & Sharif, 2010a, 2011a). An overview of GA technique to optimize the surface roughness in milling process and previous work of machining optimizing problem for surface roughness can be found in Zain, Haron, and Sharif (2008).

2.1. GA methodology

The GA technique is based on the natural process of evolution to solve optimization and search problems. There are three main operators in GA which are reproduction, crossover and mutation. To apply GA in optimization of machining process parameters, the process parameters are encoded as genes by binary encoding. The basic structure of GA optimization methodology is depicted in Fig. 2. It is important for the researcher to choose suitable GA parameters apart from weighing factors and constraints in order for the algorithm to perform efficiently. The steps to apply GA in optimization of machining are as follows (Wang & Jawahir, 2004).

(i) The process parameters are encoded as genes by binary encoding.
(ii) A set of genes is combined together to form a chromosome, which is used to perform those basic mechanisms in the GA, such as crossover and mutation.
(iii) Crossover is the operation to exchange some part of two chromosomes to generate new offspring, which is important when exploring the whole search space rapidly.
(iv) Mutation is applied after crossover to provide a small randomness to the new chromosome.
(v) To evaluate each individual or chromosome, the encoded process parameters are decoded from the chromosome and are used to predict machining performance measures.
(vi) The fitness or objective function is a function needed in the optimization process and the selection of the next generation in the GA.
(vii) After a number of iterations of the GA, optimal results of process parameters are obtained by comparison values of objective functions among all individuals.

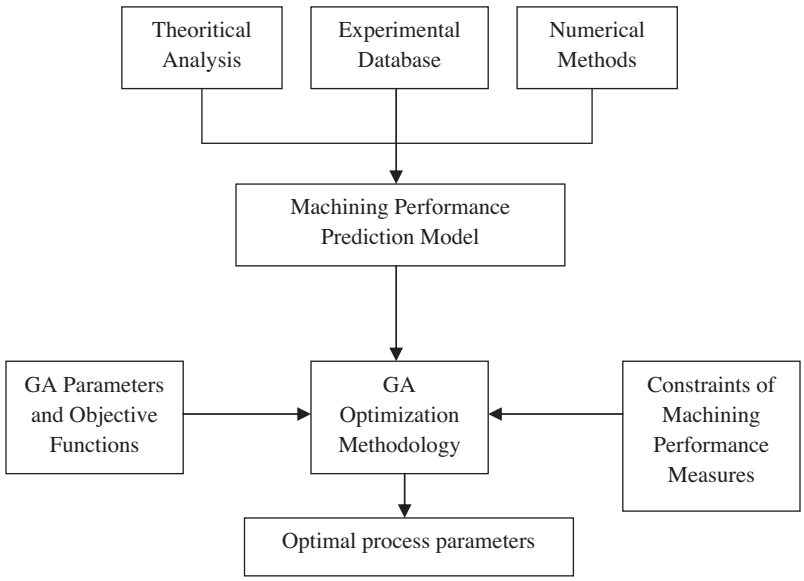

Fig. 2. GA optimization methodology (Wang & Jawahir, 2004).

2.2. Application of GA

GA optimization technique was used by Palanisamy, Rajendran, and Shanmugasundaram (2007) to find the most optimal process parameters of end milling machining such as cutting speed, depth of cut and feed rate. The objective function considered in this study was machining time. By using GA, the results showed a fast convergence and the estimated surface roughness value of 0.71 μm. From the experimental results also, the optimal process parameters have given a MRR of 6.0×10^3 mm^3/min with less amplitude of vibration at the work piece support 1.66 μm maximum displacement. The authors concluded that the optimized process parameters are capable of machining the work piece more efficiently with better surface finish. The optimization of turning, facing and undercutting process parameters using GA was considered by Saravanan and Janakiraman (2007). The objective of the research is to find the minimum machining time of the machining operations by optimizing process parameters such as cutting speed and feed rate. The GA parameters is set with the following values where sample size = 30, crossover = 0.6, mutation = 0.05 and number of generations = 100. The experimental results showed that GA reduced machining time of 5.75 s per component with 19.2% reduction of machining time in the study. A modified GA (MGA) has been proposed by Sankar, Asokan, Saravanan, Kumanan, and Prabhaharan (2007) to optimize the process parameters of multi-pass turning, facing and drilling operation. The research is divided into two different modules where the first module focusing more on multi-pass turning operations. In the second module, three machining operations such as turning, facing and drilling were used to find the optimal of average unit cost. The results of both modules have been compared with other traditional and non-traditional techniques, such as float encoded GA (FEGA), SA, ACO, Hill Climbing (HC) and Newton's method (NM). From the experimental results, it showed that MGA technique outperforms other techniques where the most optimal average cost unit has been found in both modules. The authors revealed that the modified genetic operators such as crossover and mutation improved the search more efficiently compared to the standard GA techniques. GA technique has been used by Prasad, Jayabal, and Natarajan (2007) to minimize the tool wear in turning operation. A mathematical model was developed using simple probabilistic considerations and design of experiments. GA optimization technique was used to optimize the process parameters of turning such as cutting speed, feed and depth of cut. The experimental results obtained the minimum tool wear of 0.244 mm with the optimal combination of process parameters cutting speed = 31.5 m/min, feed = 0.3 mm/rev and depth of cut = 0.5 mm in the 33th generation with the population size of 20. The results of the experiments are compared with traditional technique like dynamic programming. Duran, Barrientos, and Consalter (2007) used non-dominated sorting GA (NSGA-II) to find the optimal process parameters of turning operations such as cutting feed and feed rate. The machining performances considered in this study were production rate and production cost. The technique of NSGA-II was employed to identify economic process parameters and to show the adaptive capability of Automated Process Planning systems. Mahapatra and Patnaik (2007) employed GA technique to optimize the process parameters of WEDM with multiple objectives such as discharge current, pulse duration, pulse frequency, wire speed, wire tension, and dielectric flow. The machining performances measured in this study were metal removal rate (MRR), surface roughness and cutting width (kerf). From the results of the experiments, the researchers suggested that the process parameters of WEDM can be adjusted to achieve improved machining performances simultaneously. In the research of Parent, Songmene, and Kenné (2007), GA optimization technique was proposed to find the optimal process parameters of end milling operation. The authors presented a generalised mathematical programming model to optimize the process parameters of end milling. Then GA was employed to find the optimal process parameters. Jain, Jain, and Deb (2007) considered four types of advanced machining process (AMP) such as ultrasonic machining (UM), abrasive jet machining (AJM), waterjet machining (WJM) and abrasive waterjet machining (AWJM). All process parameters were optimized using GA techniques with the objective to maximize the MRR value. According to the researcher, real coded GA was employed because traditional methods were found to be unsuitable to solve the problems. In Jain and Jain (2007), the process parameters of electro-chemical machining such as tool feed rate, electrolyte flow velocity, and applied voltage were optimized using real coded GA. The objective of the research is to minimize geometrical inaccuracy subjected to temperature, choking, and passivity constraints. The results were compared with the past work and showed an improvement in terms of geometrical

accuracy. Lee, Nam, Choi, Kang, and Ryu (2007) presented an approach to optimize process parameters of milling machining according to the required MRR. In the study, the researcher employed 2-staged artificial neural network (ANN) as the objective function for the prediction of model surface roughness. GA was used to optimize the problem with the additional surface quality criterion. From the experimental results, the optimized machining conditions can be selected to obtain the high-quality surface within allowable reliability while maintaining a high-quality surface, under the given desired MRR. In Singh and Rao (2007) the effect of the tool geometry (effective rake angle and nose radius) and process parameters (cutting speed and feed) on the surface finish during the hard turning of the bearing steel has been investigated. In the experiments, RSM technique was used to develop first- and second-order mathematical models. Then the predicted surface roughness model was optimized by GA. The results showed that GA gives optimal values of surface roughness and their respective optimal conditions.

In Gao, Zhang, Su, and Zhang (2008), parameter optimization model was developed using both ANN and GA to optimize the process parameters of EDM such as current, pulse on time and pulse off time. An ANN model which adapts Levenberg–Marquardt algorithm has been set up to represent the relationship between MRR and process parameters. Then, GA was used to optimize the process parameters. After 250 generations, the results showed that the model is shown to be effective, and MRR is improved using optimized machining parameters. MRR values are 78.0370 mm^3/min, where current, pulse on time and pulse off time are 18 A, 416 μs, 59 μs, respectively. Mohanasundararaju, Sivasubramanian, and Alagumurthi (2008) employed two techniques, non linear programming and GA, to optimize the process parameters of grinding machining. The process parameters such as wheel speed, work speed, traverse speed, in feed, dress depth and dressing lead using Box–Behenken design matrix with six central points were considered to give a desired surface finish and dimensional accuracy. RSM technique was used to develop a second-order mathematical model. The process parameters optimization of turn milling operation has been investigated by Savas and Ozay (2008). In the experiments, the effects of process parameters on the surface roughness were optimized using GA optimization technique. The process parameters considered were depth of cut, workpiece speed, tool speed and feed rate. The optimal surface roughness for the process of tangential turn-milling was determined according to the process parameters.

In Zhang and Chen (2009), GA was used for the optimization of milling process parameters to enhance tool life and reduce processing costs. The four optimized process parameters were cutter speed, feed rate, milling depth and milling width. A mathematical model was developed based on mathematical formula and production cost process. The results showed that by using GA, the most optimized milling parameters was obtained and the most optimal tool life = 79.7852 min and processing cost = 1.438 ¥. The authors concluded that GA is easy to use and can improve the tool life and reduce processing costs.

Sultana and Dhar (2010) considered the machining process parameters of turning operation such as feed rate, pressure, flow rate and high pressure coolant to improve machining performances such as cutting temperature, chip reduction co-efficient and surface roughness. A predictive model was carried out using RSM, and multiobjective GA was used for the optimization. The results show that machining performance can be estimated by the predictive models. Yongzhi, Jun, Xiuli, and Xing (2010) used GA to optimize process parameters of high speed milling such as axial depth-of-cut, radial depth-of-cut and helical angle. A predictive model was developed using a full-factorial experimental design and multi-linear regression technology. The result shows that it is possible to select optimum for obtaining minimum cutting force and reasonably good material removal rate (MRR). In Pasam et al. (2010), eight machining process parameters of wire electrical discharge machining (WEDM) were used such as ignition pulse current, short pulse duration, time between two pulses, servo speed, servo reference voltage, injection pressure, wire speed and wire tension to find the minimum surface roughness. Taguchi technique was used to learn the behaviour of machining process parameters and regression analysis was developed to establish relationship between control parameters and surface finish. GA was used to predict the optimal surface roughness. The optimal values of machining process parameters at level for the selected range and workpiece material are obtained. Ansalam Raj and Narayanan Namboothiri (2010) proposed an improved GA, labelled as IGA. It was used to optimize machining process parameters such as feed, speed and depth of cut on surface roughness in dry turning machine. The authors noted that the main advantage of the IGA approach is that the "curse of dimensionality" and a local optimal trap inherent in mathematical programming methods can be simultaneously overcome. The IGA equipped with an improved evolutionary direction operator and a migration operation can efficiently search and actively explore solutions. The proposed IGA is more effective and applies the realistic machining problem more efficiently than the conventional GA. The research by Xu, Zhu, Wu, Zang, and Zuo (2010) was carried out to optimize process parameter of milling titanium alloy. The machining performances include cutting force, tool life and surface roughness. GA was used to find the optimal milling process parameters for the maximum metal removal rate of titanium alloy. The optimization results showed that the optimization system can improve the productivity of milling Ti6Al4V. In Alam, Nurul Amin, Patwari, and Konneh (2010), machining process parameters of NC milling such as speed, feed rate, and depth of cut were used to predict surface roughness. In the paper, quadratic prediction model was coupled with GA to optimize the machining process parameters for the minimum surface roughness. Saffar and Razfar (2010) presented a 3D simulation system to predict cutting forces during end milling operation. GA was employed to optimize the machining process parameters with the objective of minimization of the tool deflection. Tool deflection is selected as the objective function, and the constraints are surface roughness and tool life. The results are compared with experimental and indicate that the optimized process parameters are capable of machining the workpiece more accurately and with better surface finish. Bharathi and Baskar (2010) used three evolutionary optimization techniques such as SA, GA and PSO to explore the optimal machining process parameters for single pass turning operation, multi-pass turning operation and surface grinding operation. The most affecting machining parameters are considered such as number of passes, cutting speed, feed, and depth of cut. The machining performances considered in this study are the production cost and the metal removal rate in turning operation. From the experiments, it was found that GA gave better results compared to SA. However, PSO has given a better result when compared to GA optimization. GA incorporated with gene repair technique were proposed by Xie and Pan (2010) to find optimal process parameters and to minimize unit production cost in multi-pass turning operation. The selected process parameters of turning were cutting speed, feed rate and depth of cut. In the study, the population and offspring are set to 200 while the crossover and mutation rates are set to 0.36 and 0.6, respectively. The algorithm stops after 400 generations. By incorporating vector constraints-encoding and gene repair method into GA, the number of infeasible individuals in the evolutionary population was greatly reduced. Computer simulation results show that the proposed algorithm is efficient in searching the optimal machining parameters, which significantly reduce the unit production cost. Del Prete, De Vitis,

and Anglani (2010) developed a prediction model for surface roughness in flat end mill operation using RSM. ANN was used to predict surface roughness and GA was employed to optimize the surface roughness model. The process parameters considered in this study were feed, depth of cut, radial engage and speed. The developed RS model was further coupled with a developed GA to find the optimal process parameters leading to the minimum surface roughness value. The predicted optimal process parameters was validated with an experimental measurements showing that GA improved the surface roughness respect to non-optimized experimental tests from 13% to 27% depending on the different examined process parameters. By coupling developed RS model with GA, the optimization methodology is effective and can be effective if the developed RS model is accurate. In Zain et al. (2010a), GA was employed to find the optimal process parameters of end milling and abrasive waterjet operation. The machining process parameters selected for end milling are cutting speed, feed rate and radial rake angle. The results of GA are capable of estimating the optimal process parameters in end milling operation compared to experimental data, regression modelling and RSM by 27%, 26% and 50%, respectively.

Gao et al. (2011) established a model of stress and temperature field on nickel-based alloy cutting by finite element modeling and dynamic numerical simulating, and combined high-speed machining test and orthogonality analysis method. The considered machining performance is cutting force and tool wear. The tool wear and cutting force prediction model has been obtained based on the process parameters of cutting speed, feed per tooth and axial depth of cut optimized by GA. According to An, Feng, and Lu (2011) machining process parameter optimization in multi-pass milling operation involves optimal selection of cutting speed, feed rate, depth of cut, and the number of passes. A non-linear mathematical model based on minimum production cost for multi-pass milling operations is presented. GA was used to find the optimal values of the machining process parameters. The method yields lower unit production costs compared with the results from the literature and machining data handbook. In An (2011), mathematical model based on the minimum production cost criterion is developed. The machining process parameters of multi-pass turning operation are selected such as speeds, feed rates and depths of cut. The constraints of the models include tool life, surface roughness, cutting force and cutting power consumption. Optimal values of machining parameters were found by GA and two other methods which are integer programming and nonlinear programming. The model generates lower unit production costs compared with the results from the literature and machining data handbook. Kilickap, Huseyinoglu, and Yardimeden (2011) employed three machining process parameters which are cutting speed, feed rate, and cutting environment is selected to find the optimal process parameters in drilling operation. A mathematical model was developed; subsequently RSM and GA were used to determine the optimal process parameters for minimizing the surface roughness. The predicted and measured results values were quite close, which indicates that the developed model can be effectively used to predict the surface roughness. In the study of Kuruvila and Ravindra (2011), process parameters of modern machining Wire-cut Electro Discharge (WEDM) were chosen such as pulse-on duration, current, pulse-off duration, bed-speed and flushing rate. Taguchi's technique and GA were used to determine parametric influence and optimal process parameters. The results confirmed the efficiency of the approach employed for optimization of process parameters. Xie and Guo (2011) proposed a new approach by combining GA with a pass enumerating method to minimize unit production cost in multipass turning. In the pass enumerating method, the number of all possible rough cuts is calculated in order to divide the whole complicated problem into several sub-problems. In applying GA to solve the problems, the bound adjustment of optimized variants method is used to represent the chromosome in order to reduce the number of infeasible individual during evolution. Computer simulation results showed that the proposed optimization approach can find the better results than other algorithms proposed previously to significantly reduce the unit production cost. In Rai et al. (2011), the prediction of optimal machining process parameters such as axial depth of cut, radial immersion, feed rate and spindle speed in multi-tool milling operation was done based on a model named GA-MPO (GA based milling parameter optimisation system). From the results, the developed system enhanced functional capabilities and gives accurate prediction compared to other models. Zeng, E, Yang, and Li (2011) built a soft-sensing model for optimizing machining process parameters such as rotate speed, speed and depth of cutting based on support vector machines. Adaptive GA was used to optimize the allowable error, positive aligned and the kernel function parameter. After being optimized 300 steps, the average relative error tended to saturation training was 4.0%; the test error was less than 2.6%; the average relative error between the Soft-sensing value for the roughness of machining surface under the numerical control. Optimization of five machining processes in abrsive waterjet (AWJ) machining which are traverse speed, waterjet pressure, standoff distance, abrasive grit size and abrasive flow rate was presented by Zain et al. (2011a). The results showed that GA found optimal process parameters that lead to much lower surface roughness value compared to SA. Also in Zain, Haron, and Sharif (2012), the authors proposed the integration of ANN and GA techniques to find optimal process parameters value (speed, feed and radial rake angle) of end milling machining that lead to minimum value of surface roughness. The experimental results showed that the minimal surface roughness value achieved was 0.139 μm and the optimal process parameters were, feed = 167.029 m/min, speed = (0.025 mm/tooth), and radial rake angle = 4.769°. The authors stated that the surface roughness value achieved was much lower about 26.8%, 25.7%, 26.1% and 49.8%, compared to the experimental, regression, ANN and RSM results, respectively. The experiments also reduced the mean surface roughness value and number of iterations about 0.61% and 23.9%, respectively compared to the conventional GA results. Table 1 summarized the latest researches in optimizing process parameters of traditional and modern machining using GA techniques.

3. Simulated annealing

SA optimization technique is based on random numbers for the evaluation of the objective function that gives global optimum solution (Bharathi & Baskar, 2010). SA was proposed by Kirkpatrick, Gelatt, and Vecchi (1983) to find the optimal global cost function that may possess several local optima (Bertsimas & Tsitsiklis, 1993; Cerny, 1985). SA technique imitates the process of gradual cooling of metals in nature. Compared to other global optimization such as GA and TS, SA is easier to put into practice and provide good solution for many combinatorial problems. The parameters of standard SA include initial temperature and decrement (cool down) factor. In Rao, Pawar, and Davim (2010b), the researchers employed SA techniques to optimize process parameters of mechanical type advanced machining and the result shows that SA outperformed the GA techniques.

3.1. SA methodology

The SA optimization flowchart is shown in Fig. 3 and the SA algorithms are as follows (Yang et al., 2009):

Table 1
Summary of recent GA techniques in optimizing machining process parameters.

No	Author/year	Process parameters	Machining process	Machining performance	Remarks
1.	Rai et al. (2011)	Axial depth of cut, radial immersion, feed rate and spindle speed	Multi-tool milling	Machining time	GA-MPO enhanced functional capabilities and gives accurate prediction compared to other models
2.	Zeng et al. (2011)	Rotate speed, speed and depth of cutting	N/A	Surface roughness	The average relative error tended to saturation training was 4.0%; the test error was less than 2.6%
3.	Gao et al. (2011)	Bonding wear, feed per tooth and axial depth of cut	High speed machining (nickel-based alloy cutting)	Cutting force tool, tool life	The influence of cutting speed on cutting force is smaller than feed per tooth and axial depth of cut
4.	An et al. (2011)	Speed, feed rate, depth of cut, and the number of passes	Multi-pass milling	Production costs	The method yields lower unit production costs compared with the results from the literature and machining data handbook
5.	An (2011)	Speed, feed rate and depth of cut	Multi-pass turning	Production costs	Lower unit production costs compared with the results from the literature and machining data handbook
6.	Kilickap et al. (2011)	Cutting speed, feed rate, and cutting environment	Drilling	Surface roughness	The developed model can be effectively used to predict the surface roughness
7.	Kuruvila and Ravindra (2011)	Pulse-on duration, current, pulse-off duration, bed-speed and flushing rate	WEDM	Dimensional error, surface roughness, volumetric MRR	The results confirm the efficiency of the approach employed for optimization of process parameters in this study
8.	Ganesan et al. (2011)	Depth of cut, cutting speed and feed	Multi-pass turning	Production time	GA and PSO have been employed to find the optimal machining parameters for the continuous profile
9.	Xie and Guo (2011)	Depth of cut, cutting speed and feed	Multi-pass turning	Production costs	The optimization approach can find the better results than other algorithms proposed previously to significantly reduce the unit costs
10.	Zain et al. (2010a)	Cutting speed, feed rate and radial rake angle	End milling	Surface roughness	GA is capable of estimating the optimal process parameters compared to experimental data, regression modelling and RSM by 27%, 26% and 50%, respectively
11.	Zain et al. (2011a)	Traverse speed, waterjet pressure, standoff distance, abrasive grit size and abrasive flow rate	AWJ	Surface roughness	The results show that GA found optimal surface roughness value compared in regression and experimental
12.	Zain et al. (2012)	Cutting speed, feed rate and radial rake angle	End milling	Surface roughness	Compared to the conventional GA, the proposed techniques showed good results where it reduced the mean value of surface roughness and number of iterations by 0.61% and 23.9%, respectively
13.	Zain et al. (2011c)	Radial rake angle, cutting speed and feed	End milling	Surface roughness	The proposed integration of SA and GA gives a lower number of iterations compared to conventional techniques of SA and GA
14.	Sultana and Dhar (2010)	Feed rate, pressure, flow rate and high pressure coolant	Turning	Cutting temperature, chip reduction co-efficient and surface roughness	The results show that machining performance can be estimated by the predictive models
15.	Yongzhi et al. (2010)	Axial depth-of-cut, radial depth-of-cut and helical angle	High speed milling	Cutting force, metal removal rate	The result shows that it is possible to select optimum for obtaining minimum cutting force and reasonably good metal removal rate
16.	Pasam et al. (2010)	Ignition pulse current, short pulse duration, time between two pulses, servo speed, servo reference voltage, injection pressure, wire speed and wire tension	WEDM	Surface roughness	The optimal values of machining process parameters at level for the selected range and workpiece material are obtained
17.	Ansalam Raj and Narayanan Namboothiri (2010)	Feed, speed and depth of cut	Dry turning	Surface roughness	The proposed IGA is more effective and applies the realistic machining problem more efficiently than does the conventional GA (CGA)
18.	Alam et al. (2010)	Speed, feed rate, and depth of cut	NC milling	Surface roughness	It is observed that cutting speed has the most significant influence on surface roughness followed by feed and depth of cut
19.	Xu et al. (2010)	Feed rate, depth of cutting, cutting width	Milling	Cutting force, tool life and machined surface roughness, metal removal rate	The optimization results show the optimization system can improve the productivity of milling Ti6Al4V
20.	Saffar and Razfar (2010)	Cutting speed, feed rate and radial rake angle	End milling	Cutting force	The obtained results indicate that the optimized parameters are capable of machining the workpiece more accurately and with better surface finish
21.	Bharathi and Baskar (2010)	Number of passes, cutting speed, feed, and depth of cut	Single pass turning multi-pass turning,	Production cost, metal removal rate	From the experiments GA did not give better results compared to PSO but not gives better results than SA in the three turning operation

Table 1 (*continued*)

No	Author/year	Process parameters	Machining process	Machining performance	Remarks
			and surface grinding		
22.	Jayabal and Natarajan (2010)	Bit diameter, spindle speed, and feed rate	Drilling	Thrust force, torque, and tool wear	minimum drill forces are obtained by operating the machine at lower drill bit diameter, higher speed, and lower feed rate
23.	Xie and Pan (2010)	Speed, feed rate and depth of cut	Multi-pass turning	Production costs	The proposed algorithm is efficient in searching the optimal machining parameters, which significantly reduce the unit production cost
24.	Del Prete et al. (2010)	Feed, depth of cut, radial engage and speed	Flat end mill	Surface roughness	GA improved the surface roughness respect to non-optimized experimental tests from 13% to 27%
25.	Maji and Pratihar (2010)	Peak current, pulse-on-time and pulse-duty-factor	EDM	Surface roughness, MRR	The optimal results found to be satisfactory and Pareto-optimal front of solutions had been obtained
26.	Zhang and Chen (2009)	Rotation speed, feed rate, depth of cutting, cutting width	Milling	Tool life, processing costs	The most optimized milling parameters were obtained
27.	Wang et al. (2009)		High speed milling	Surface roughness, MRR	The authors stated that the MRR can be achieved in the certain range of surface roughness by choosing the right cutting parameters
28.	Gao et al. (2008)	Current, pulse on time and pulse off time	EDM	MRR	MRR is improved by using optimized parameters
29.	Mohanasundararaju et al. (2008)	Wheel speed, work speed, traverse speed, in feed, dress depth and dressing lead	Grinding	Surface roughness	The two optimization approaches were used namely non-linear programming and GA (GA)
30.	Savas and Ozay (2008)	Depth of cut, workpiece speed, tool speed and feed rate	Turn milling	Surface roughness	the errors in measurement regions are smaller than 7% and greater than 2%
31.	Saravanan and Janakiraman (2007)	Cutting speed and feed rate	Turning, facing and undercutting	Machining Time	GA technique helps the production engineers by maximizing the production rate and minimizing the production cost
32.	Sankar et al. (2007)	Cutting speed, feed rate and depth of cut	Multi-pass turning, facing and drilling	Production costs	The modification in the genetic operators improves the search in a more effective way than the classical genetic algorithm
33.	Prasad et al. (2007)	Cutting speed, feed rate and depth of cut	Turning	Tool wear	The results of the experiments are compared with traditional technique like dynamic programming
34.	Duran et al. (2007)	Cutting feed and feed rate	Turning	Production rate and production cost	The Pareto front margins correspond to or are comparable to the limits of the high efficiency cutting range
35.	Mahapatra and Patnaik (2007)	Discharge current, pulse duration, pulse frequency, wire speed, wire tension, and dielectric flow.	WEDM	MRR, surface roughness and cutting width (kerf).	The process parameters of WEDM can be adjusted to achieve improved machining performances simultaneously
36.	Parent et al. (2007)	Cutting speed, feed rate and depth of cut	End milling	Production costs	GA was proposed to find optimal machining process parameters
37.	Jain et al. (2007)	**UM** Amplitude of vibration, frequency of vibration, mean diameter of abrasive grains, volumetric concentration of abrasive particles in slurry, static feed force **AJM** Mass flow rate of abrasive particles, mean radius of abrasive particles, velocity of abrasive particles, **WJM** Water jet pressure at the nozzle exit, diameter of water jet nozzle, traverse rate of the nozzle **AWJM** Water jet pressure at the nozzle exit, diameter of abrasive-water jet nozzle, traverse or feed rate of the nozzle, mass flow rate of water, mass flow rate of abrasives	UM, AJM, WJM, AWJM	MRR	GA was used for solving the formulated optimization models
38.	Palanisamy et al. (2007)	Cutting speed, depth of cut and feed rate	End milling	Machining time	The optimized process parameters are capable of machining the work piece more efficiently with better surface finish
39.	Jain and Jain (2007)	Tool feed rate, electrolyte flow velocity	Electro-chemical machining	Geometrical accuracy	The results is compared with the past work and showed an improvement in terms of geometrical accuracy
40.	Lee et al. (2007)	Rotation speed, feed rate, depth of cutting, cutting width	Milling	Surface roughness, MRR	It has been investigated that optimized machining conditions can be selected to obtain the high-quality surface within allowable reliability while maintaining a high-quality surface, under the given desired MRR
41.	Singh and Rao (2007)	Cutting speed and feed	Hard turning	Surface roughness	The GA gives minimum values of surface roughness and their respective optimal conditions

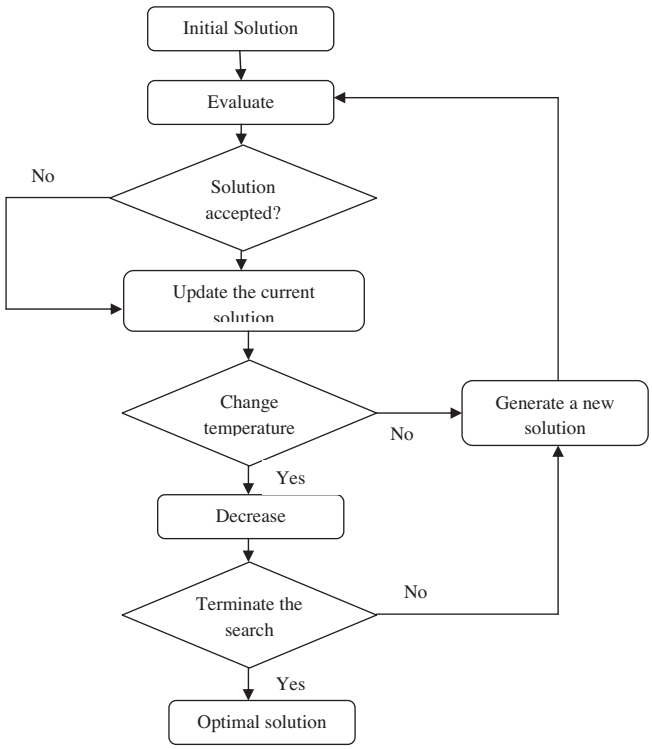

Fig. 3. SA optimization flowcharts (Zain et al., 2010b).

(i) Choose a randomly generated initial point X^0, a termination temperature T_{low}. Also set number of iterations (N) to be performed at a particular temperature and iteration counter $t = 0$.
(ii) Evaluate the value of objective function $E_1 = f(X^t)$.
(iii) Calculate a neighborhood point X^{t+1} using random perturbation and evaluate objective function at X^{t+1} as $E_2 = f(X^{t+1})$.
(iv) Calculate $\Delta E = E_2 - E_1$.
(v) If $\Delta E<0$, accept the point. That is $X^t = X^{t+1}$ and $E_1 = E_2$. Set $t = t + 1$ and go to step (vi)
(vi) If $\Delta E \geqslant 0$, create random number r in the range (0, 1) and check whether $r \leqslant \exp(-\Delta E/T)$. If satisfied then set $t = t + 1$ and go to step-6. Else begin with new initial point Xt and go to step-3.
(vii) If $t > N$ go to step-7.
(viii) Reduce the temperature periodically by a factor k_1 according to $T = k_1 T$ and go to step (iii)
(ix) If $T \leqslant T_{low}$ then terminate the process.

In Chen, Lin, Yang, and Tsai (2010), to define fitness function of $S(x)$, the formulation of using SA techniques is defined by the following formula:

Minimize,

$$S(x) = \sum_{i=1}^{k} (y_{ti} - y_{pi})^2 \tag{1}$$

Subject to,

$$\mathrm{LCL}_i \leqslant y_{pi} \leqslant \mathrm{UCL}_i \tag{2}$$

$$\mathrm{LCL} = \mu - n\varepsilon; \quad n = 1, 2, \ldots, N \tag{3}$$

$$\mathrm{UCL} = \mu + n\varepsilon; \quad n = 1, 2, \ldots, N \tag{4}$$

where, is the x is the process parameters, K is the total number of response which is nominal the best type and has certain target, y_t is the predicted value of i response that is a nominal the best type response, y_p is the certain target value, UCL_i is the upper control limit of i response, LCL_i is the lower control limit of i response, μ: mean value of experimental data, and ε is the standard deviation of experimental data.

3.2. Application of SA

Kolahan and Abachizadeh (2008) developed SA algorithm to optimize machining process parameters in turning operation on cylindrical workpieces. Three process parameters of turning operation were chosen which are cutting speed, feed rate and cutting depth. The machining performance considered in this study is to minimize the machining cost. The optimized process parameters of cutting speed, feed rate and cutting depth are 145 m/min, 0.25 mm/rev, and 2.75 mm, respectively. The total cost achieved is $37.58. The computational results clearly show that the proposed optimization procedure has considerably improved total operation cost by optimally determining machining parameters. In Satishkumar and Asokan (2008), process parameters of CNC multi-tool drilling system were optimized to minimize production cost and incorporate various technological and machine tool constraints. Three evolutionary techniques were considered such as GA, SA and ACO to find the optimal process parameters of the machining operation.

Yang (2009) proposed an optimization methodology for the selection of best process parameters in electro-discharge machining. There are four process parameters selected for EDM machining which are discharge current, source voltage, pulse-on time and pulse-off time. Process parameters were optimized by SA technique to maximize the MRR on top of minimize the surface roughness. The optimal surface roughness achieved is 2.07 μm and the maximum value of MRR is 54.93 g/h. Kolahan and Khajavi (2009) evaluated the influences of AWJ process parameters such as nozzle diameter, jet traverse rate, jet pressure and abrasive flow rate in cutting 6063-T6 aluminum alloy. The Taguchi method and regression modeling were used in order to establish the relationships between input and output parameters. SA was used to optimize the AWJ process parameters. The settings of SA parameters in this study are as follows: initial temperature (T_0) = 250, cooling rate (α) = 0.98 and the termination criteria = 500 iterations. The objective is to determine a suitable set of process parameters that can produce a desired depth of cut. The results confirmed the effectiveness of the proposed model and optimization procedure where all the process parameters deviate from their desired values by less than 0.5%.

In the study by Zain, Haron, and Sharif (2010b), three parameters of end milling were considered for minimizing surface roughness. From the experiments, it was recommended that process parameters should be set at the highest cutting speed, lowest feed and highest radial rake angle in order to achieve the minimum surface roughness of 0.1385 μm. The minimum surface roughness was much lower than the experimental sample data, regression modelling and RSM technique by 27%, 26% and 50%, respectively. Also in Zain et al. (2011a) five parameters of AWJ in cutting 6063-T6 aluminum alloy such as traverse speed, waterjet pressure, standoff distance, abrasive grit size and abrasive flow rate were selected to find the optimal surface roughness. SA was used to optimize the AWJ process parameters and the computational results prove the effectiveness of the proposed model and optimization procedure.

The study of Chen et al. (2010) analyzed WEDM process parameters during manufacture of pure tungsten profiles. The pulse on time, the pulse off time, arc off time, the servo voltage, the wire feed rate, the wire tension and the water pressure were selected as the WEDM process parameters. Three considered machining performances are the cutting velocity, surface roughness and roughness maximum. Integrate BPNN/SAA approaches was proposed and SAA techniques was used to find the most optimal

process parameters. The estimated optimal process parameters are: pulse on time of 0.42 μs, pulse off time of =12.15 μs, arc off time = 13.73 μs, servo voltage = 45.17 V, wire feed rate = 10.32 m/mm, wire tension = 1751.07 gf, and water pressure = 15.21 kgf/cm^2. The predicted machining performance cutting velocity = 7.8558 m/min, surface roughness = 1.1786 μm roughness maximum = 10.7873 μm. Rao and Pawar (2010b) optimized the process parameters of multi-pass milling operation such as the number of passes, depth of cut, cutting speed and feed to minimize the production time (i.e., maximization of production rate). SA was employed to find the optimal process parameters. The results of SA were compared with the previously published results obtained by using other optimization techniques, ABC and PSO optimization.

Zain, Haron, and Sharif (2011b) proposed the integration of ANN and SA techniques to optimize the process parameters of AWJ such as traverse speed, waterjet pressure, standoff distance, abrasive grit size and abrasive flow rate. The machining performance measured in this study was surface roughness. In this study, the integration of ANN and SA techniques were divided into two categories namely ANN-SA type1 and ANN-SA type 2. From the experimental results, the proposed integrations system has been successfully optimized process parameters of AWJ and gave a minimal value of surface roughness = 1.523 μm. Also in Zain, Haron, and Sharif (2011c), two soft computing techniques which are SA and GA were integrated to find the optimal process parameters of end milling machining that lead to the minimum value of surface roughness. The integration of SA-GA was also divided into two categories which are SA-GA type1 and SA-GA type2. The results of the experiments showed that the proposed technique was effective in optimizing the process parameters of end milling machining and the time for searching the optimal solution can also be made faster. The latest researches in optimizing process parameters of traditional and modern machining using SA techniques is shown in Table 2.

4. Particle swarm optimization

PSO technique was introduced by Kennedy and Eberhart (1995) to solve continuous optimization problems (Li, Yao, Gao, Liu, & Yuan, 2008). The swarm is composed of volume-less particles with stochastic velocities, each of which represents a feasible solution. The algorithm finds the optimal solution through moving the particles in the solution space.

4.1. PSO methodology

The implementation of PSO is very simple and needs only a few lines programming code. The flow chart of the PSO algorithm is depicted in Fig. 5. It requires uncomplicated mathematical operators; therefore it is computationally economical in terms of both memory requirements and speed. PSO has features of both GA and evolution strategies (Zŭperl, Cŭs, & Gecevska, 2007). The PSO framework for process parameter optimization is depicted in Fig. 4. The steps of optimizing process parameters of milling operation using PSO was given by Zŭperl, Cŭs, and Gecevska (2007) as follows.

(i) Generation and initialization of an array of 50 particles with random positions and velocities. Velocity vector has two dimensions, feed rate and spindle speed.
(ii) Evaluation of objective (cutting force surface) function for each particle.
(iii) The cutting force values are calculated for new positions of each particle. If a better position is achieved by particle, the pbest value is replaced by the current value.
(iv) Determination if the particle has found the maximal force in the population. If the new gbest value is better than previous gbest value, the gbest value is replaced by the current gbest value and stored. The result of optimization is vector gbest (feedrate, spindle speed).

Table 2
Summary of recent SA techniques in optimizing machining process parameters.

No.	Author/year	Process parameters	Machining process	Machining performance	Remarks
1.	Zain et al. (2010b)	Radial rake angle, cutting speed and feed	End Milling	Surface roughness	The minimum surface roughness was much lower compared to experimental, regression and RSM
2.	Zain et al. (2011a)	Traverse speed, waterjet pressure, standoff distance, abrasive grit size and abrasive flow rate	AWJ	Surface roughness	The optimal surface roughness value in SA is more less compared to experimental, regression, and GA
3.	Zain et al. (2011b)	Traverse speed, waterjet pressure, standoff distance, abrasive grit size and abrasive flow rate	AWJ	Surface roughness	The results showed that an optimal values of surface roughness and a lower number of iterations are obtained using the proposed techniques
4.	Zain et al. (2011c)	Radial rake angle, cutting speed and feed	End milling	Surface roughness	The proposed integration of SA and GA gives a lower number of iterations compared to conventional techniques of SA and GA
5.	Bharathi and Baskar (2010)	Number of passes, cutting speed, feed, and depth of cut	Single pass turning multi-pass turning, and surface grinding	Production cost, metal removal rate	From the results SA did not give better results compared to PSO and GA in the three turning operation
6.	Yang et.al., (2009)	Discharge current, source voltage, pulse-on time and pulse-off time	EDM	Surface roughness, metal removal rate	The optimal surface roughness achieved is 2.07 and the maximum value of MRR is 54.93
7.	(Chen et al., 2010)	Pulse on time, the pulse off time, arc off time, the servo voltage, the wire feed rate, the wire tension and the water pressure	WEDM	Cutting velocity, surface roughness, metal removal rate	From the results and conformation of experiments, BPNN/SAA method is effective tool for the optimization of WEDM process parameters
8.	Rao and Pawar (2010b)	Number of passes, depth of cut, cutting speed and feed	Multi-pass milling	Production time	The results are compared with the previously published results obtained by using other optimization techniques
9.	Kolahan and Khajavi (2009)	Nozzle diameter, jet traverse rate, jet pressure and abrasive flow rate	AWJ	Depth of cut	SA algorithms provide an effective and speedy optimization technique
10.	Kolahan and Abachizadeh (2008)	Cutting speed, feed rate and cutting depth	Turning	Machining cost	The results improved total operation cost
11.	Satishkumar and Asokan (2008)	Cutting speed, feed rate, and cutting environment	Multi-tool drilling	Production cost	GA and ACO also considered in this study in optimizing the machining process parameters

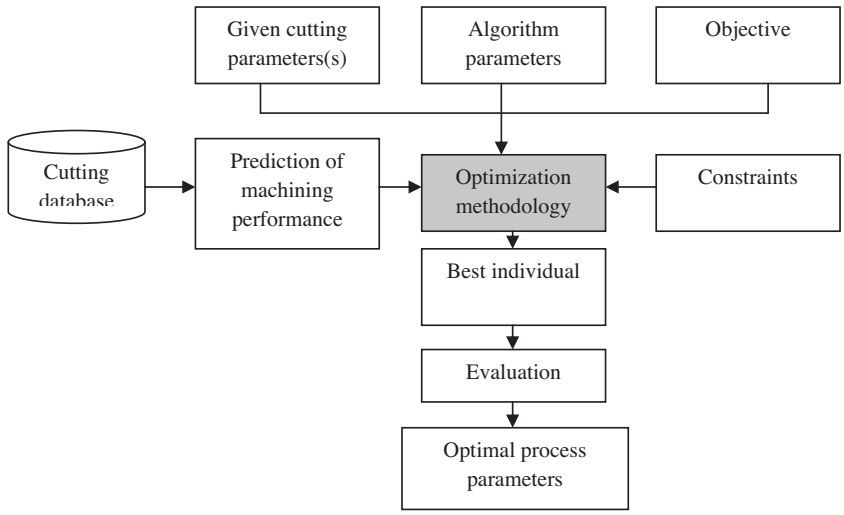

Fig. 4. Framework PSO process parameter optimization methodology (Li et al., 2008).

(v) Computation of particles' new velocity.
(vi) Update particle's position by moving towards maximal cutting force.
(vii) Steps (i) and (ii) are repeated until the iteration number reaches a predetermined iteration.

4.2. Application of PSO

Zŭperl, Cŭs, and Gecevska (2007) employed PSO to optimize process parameters of milling machining. A predictive model was developed using ANN to predict the cutting forces during machining and PSO was used later to obtain optimal process parameters of milling machining such as cutting speed and feed rates. The results were compared with other evolutionary techniques such as GA and SA and proved that the proposed technique improved the quality of the solution while speeding up the convergence process. A new technique has been proposed by Huang, Li, and Lin (2007) by using the combination of wavelet neural network (WNN) algorithm and modified PSO for solving tool wear detection and estimation. By using the Daubechies-wavelet, the cutting power signal is decomposed into approximation and details. The energy and square-error of the signals in the detail levels is used as characters which indicating tool wear, the characters are input to the trained WNN to estimate the tool wear. The results of the experiments were compared with BP neutral network, conventional WNN and GA-based WNN. The results showed a faster convergence and more accurate estimation of tool wear.

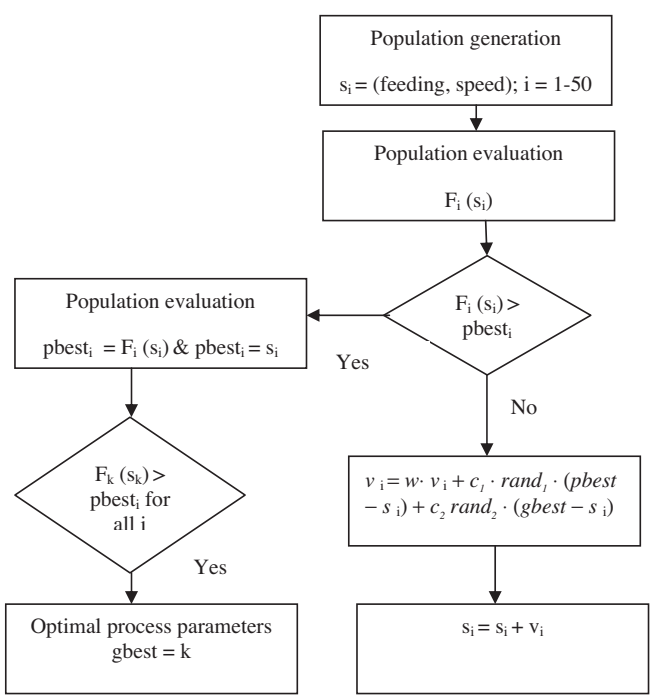

Fig. 5. PSO optimization for optimal process parameters (Zŭperl, Cŭs, & Gecevska, 2007).

According to Rao, Pawar, and Shankar (2008), process parameters of electrochemical machining (ECM) such as the tool feed rate, electrolyte flow velocity, and applied voltage play a significant role in optimizing the measures of process performance. PSO was used to find the optimal combination of process parameters for an ECM operation. There are three machining performance measured which includes dimensional accuracy, tool life, and the MRR. The results of the proposed algorithm are compared with the previously published results obtained by using other optimization techniques. The process parameters of milling operation such as spindle speed and feed rate were considered to be optimized in the study of Li et al. (2008). The considered machining performances were cutting force, tool-life, surface roughness and cutting power. An algorithm for process parameters optimization known as cutting parameters optimization (CPO) was introduced and PSO technique was employed to optimize the process parameters. From the experimental results, the authors concluded that PSO in optimizing process parameters can converge quickly to a consistent combination of spindle speed and feed rate. An application was build in Duran, Rodriguez, and Consalter (2008) to select suitable cutting tool geometry in a given combination of material work piece and cutting tool material. PSO was employed to find the optimal cutting tool geometry and evaluates a selected number of individuals (that represent a set of feasible tool angle) until a termination criteria is satisfied. In the experiments, a range of simulations were carried out to confirm the performance of the algorithm and to show the usefulness of the suggested approach. Chen and Li (2008) proposed an improved PSO with opposition mutation (OMPSO) to select satisfied process parameter (depth of cut, feed rate, grit size) of grinding process. According to the researcher, OMPSO has the same tuning parameters as PSO and easy to use. The experiment result was compared to other

evolutionary techniques such as GA, PSO and landscape adaptive PSO (LAPSO). It was obtained that the proposed technique was effective to solve grinding process optimization problem. The optimization of process parameters for constant cutting force was discussed based-on virtual machining by Zhao, Li, Yao, and Liu (2008). PSO was employed to find the optimal process parameters (spindle speed and feed rate). The framework of virtual machining based cutting parameters optimization was established. Then two controlled experiments were conducted to demonstrate the effectiveness of cutting parameters optimization both with physical cutting and computer simulation. The results of experiment showed that machining process with constant cutting force can be achieved via cutting parameters optimization based on virtual machining. Tang, Landers, and Balakrishnan (2008) investigated two-tool parallel turning (single pass and multipass) process parameters optimization problem. PSO was employed to determine optimal machining time. The results showed that the proposed technique performed better than exhaustive search algorithm in terms of machining time and required computational time.

Optimization of process parameters in turning operation was studied by Xi and Liao (2009). There are three objectives control parameters, which are machining time, machining accuracy and machining cost. The model was established using multiple targets nonlinear programming model. The process parameters were optimized using PSO. From the experimental results, the researchers found the optimal process parameters (cutting speed and feed rate) value is much smaller than the value calculated by the experience of the objective function value. The optimized cutting parameters values are better meet the user's optimization goals than obtained from the experience or manuals on the recommended values and more reference value. PSO was used in the research by Escamilla, Perez, Torres, Zambrano, and Gonzalez (2009) to find optimal process parameters of the titanium's machining process. For the modelling and prediction of the process outputs, ANN network was employed for Vertical Machining Center Bridgeport VMC 760. The machining the tool was an end mill coated with Aluminium Titanium Nitride (AlTiN). The obtained surface roughness value was 0.68 μm and the optimal process parameters values of speed, feed and depth of cut is 2798 m/min, 425 mm/rev and 0.5 mm, respectively. From the results of ANN modelling and PSO optimization, it can be successfully applied to multi-objective optimization of titanium's machining process. Modeling and optimizing process parameters in pulsed laser micromachining is the main focused in Ciurana, Arias, and Ozel (2009). Selection of process operational parameters is highly critical for successful laser micromachining. The relation between process parameters and quality characteristics has been modeled with ANN. Predictions with ANNs have been compared with experimental work. Multiobjective PSO of process parameters for minimum surface roughness and minimum volume error is carried out. This result shows that the proposed model and swarm optimization approach are suitable to identify optimum process settings. In the research by Prakasvudhisarn, Kunnapapdeelert, and Yenradee (2009), process parameters of CNC end milling were selected such as feed rate, spindle speed, and depth of cut to find the minimum surface roughness. Support vector machine (SVM) was proposed to capture characteristics of roughness and its factors. PSO technique is then employed to find the combination of optimal process parameters. The results showed that cooperation between both techniques can achieve the desired surface roughness and also maximize productivity simultaneously. Srinivas, Giri, and Yang (2009) proposed a methodology for selecting optimum machining parameters in multi-pass turning using PSO. The considered machining performances are production cost and machining time. PSO was implemented to obtain the set of cutting parameters that minimize unit production cost subject to practical constraints. The dynamic objective function approach adopted in the paper resolves a complex, multi-constrained, nonlinear turning model into a single, unconstrained objective problem. The best solution in each generation is obtained by comparing the unit production cost and the total non-dimensional constraint violation among all of the particles.

Razfar, Asadnia, Haghshenas, and Farahnakian (2010) proposed a PSO-based neural network to create a predictive model for the surface roughness level that is based on experimental data collected on e face milling X20Cr13 stainless steel. The optimization problem is then solved using a PSO-based neural network for optimization system (PSONNOS). A good agreement is observed between the predicted surface roughness values and those obtained in experimental measurements performed using the predicted optimal machine settings. The PSONNOS is compared to the GA optimized neural network system (GONNS). PSO was used by Zheng and Ponnambalam (2010) to optimize the multipass turning process which has rough machining and then a finish machining. The considered objective function is minimization of unit production cost. The performance is evaluated by comparing results of PSO with GA and SA that were reported by earlier researchers. Bharathi and Baskar (2010) used three evolutionary optimization techniques such as SA, GA and PSO to explore the optimal machining process parameters for single pass turning operation, multi-pass turning operation, and surface grinding operation. The most affecting machining parameters are considered such as number of passes, cutting speed, feed, and depth of cut. The machining performances considered in this study are the production cost and the metal removal rate. The result of PSO is 4.7% and 1% better than GA and SA, respectively. In multi-pass turning operation, the result of PSO is 12.5% and 19.8% better than GA and SA, respectively. In grinding operation, the result of PSO is 6.2% and 1% better than GA and SA, respectively. PSO also gave better results compared to GA and SA in the three turning operations.

The machining performance considered in Bharathi and Baskar (2011) are machining time and surface roughness. CNC turning machine was employed to conduct experiments on brass, aluminium, copper, and mild steel. PSO has been used to find the optimal machining parameters for minimizing machining time subjected to desired surface roughness. Physical constraints for both experiment and theoretical approach are cutting speed, feed, depth of cut, and surface roughness. It is observed that the machining time and surface roughness based on PSO are nearly same as that of the values obtained based on confirmation experiments; hence, it is found that PSO is capable of selecting appropriate machining parameters for turning operation. In the research by Farahnakian, Razfar, Moghri, and Asadnia (2011), the effect of process parameters of high speed steel end mill such as spindle speed and feed rate are considered. Nanoclay (NC) content on machinability properties of polyamide-6/nanoclay (PA-6/NC) nanocomposites was studied for modeling cutting forces and surface roughness by using PSO-based neural network (PSONN). The results indicate that the nanoclay content on PA-6 significantly decreases the cutting forces, but does not have a considerable effect on surface roughness. The obtained results for modeling cutting forces and surface roughness also showed a remarkable training capacity of the proposed algorithm compared to the conventional neural network. Yang, Guo, and Liao (2011a) proposed a methodology, fuzzy PSO (FPSO) algorithm to distribute the total stock removal in each of the rough passes and the final finish pass which based on fuzzy velocity updating strategy to optimize the machining parameters implemented for multi-pass face milling. The optimum value of machining parameters including number of passes, depth of cut in each pass, speed, and feed are obtained to achieve minimum production cost. The proposed methodology for distribution of the total stock removal in each of passes is effective, and the proposed FPSO algorithm does not have any difficulty in converging towards the true

optimum. From the given results, the proposed schemes may be a promising tool for the optimization of machining process parameters. Also in Yang, Guo, and Liao (2011b), the researchers proposed fuzzy global and personal best-mechanism-based multi-objective PSO (F-MOPSO) to optimize the machining parameters. The proposed algorithm was used to optimize the machining parameters is developed to solve such a multi-objective optimization problem in optimization of multi-pass face milling operation. It was found that the F-MOPSO does not have any difficulty in achieving well-spread Pareto optimal solutions with good convergence to true Pareto optimal front for multi-objective optimization problems. Costa, Celano, and Fichera (2011) used hybrid PSO for minimizing the production cost associated with multi-pass turning problems. The proposed optimization technique consists of a PSO-based framework wherein a properly embedded SA, namely an SA-based local search, aims both to enhance the PSO search mechanism and to move the PSO away from being closed within local optima. The used process parameters are cutting speed, feed rate and depth of cut. Five different test cases based on the multi-pass turning of a bar stock have been used for comparing the performance of the proposed technique with other existing methods. In Ganesan, Mohankumar, Ganesan, and Ramesh Kumar (2011), the machining parameters in multipass turning such as depth of cut, cutting speed and feed are considered. These process parameters were optimized using GA and PSO for minimization of production time. In GA the combination of optimal process parameters speed, feed and depth of cut achieved is 2185.714 m/min, 0.22 mm/rev and 0.87 mm, respectively with minimum production time = 3.131 min. In PSO, combination of optimal process parameters speed, feed and depth of cut achieved is 3500.000000 m/min, 0.367393 mm/rev and 0.010000 mm, respectively with minimum production time = 0.000180 min. It was found that PSO gave better results compared to GA. Table 3 summarized the latest researches in optimizing process parameters of traditional and modern machining using PSO techniques.

5. Artificial bee colony optimization

ABC is the recent swarm-based algorithm that mimics the foraging behaviour of swarm honey bee. Similar to the concept of ACO and PSO, this exploration algorithm is capable of tracing good quality of solutions.

5.1. ABC methodology

There are three control parameters that perform significant role in the ABC which is the number of colony, the value of limit and the maximum loop for searching. The abilities of ABC algorithm have been discussed by Rao et al. (2008), Karaboga (2009), Benala, Jampala, Villa, and Konathala (2009), Akay and Karaboga (2010), Karaboga and Akay (2009) Rao and Pawar (2010b) and Akay and Karaboga (2009). The flowchart of ABC is shown in Fig. 6. The detailed pseudocode to solve the optimization is as follows (Karaboga & Akay, 2009):

(i) initialize the population of solutions $x_{i,j}$
(ii) evaluate the population
(iii) cycle = 1
(iv) repeat
(v) produce new solutions (food source positions) $v_{i,j}$ in the neighbourhood of $x_{i,j}$ for the employed bees using the formula $v_{i,j} = x_{i,j} + \Phi ij(x_{i,j} - x_{k,j})$ (k is a solution in the neighbourhood of i, Φ is a random number in the range $[-1, 1]$) and evaluate them
(vi) apply the greedy selection process between x_i and v_i
(vii) Calculate the probability values P_i for the solutions x_i by means of their fitness values using Eq. (5):

$$P_i = \frac{fit_i}{\sum_{i=1}^{SN} fit_i} \tag{5}$$

In order to calculate the fitness values of solutions, the following equation is employed (6):

$$fit_i = \begin{cases} \frac{1}{1+f_i}, & \text{if } f_i \geqslant 0 \\ 1 + abs\, f(i), & \text{if } f_i < 0 \end{cases} \tag{6}$$

(viii) Normalize P_i values into [0,1]
(ix) produce the new solutions (new positions) v_i for the onlookers from the solutions x_i, selected depending on P_i, and evaluate them.
(x) apply the greedy selection process for the onlookers between x_i and v_i.
(xi) determine the abandoned solution (source), if exists, and replace it with a new randomly produced solution x_i for the scout using Eq. (7):

$$x_{ij} = \min_j + \text{rand}(0,1) * (\max_j - \min_j) \tag{7}$$

(xii) Memorize the best food source position (solution) achieved so far
(xiii) cycle = cycle + 1
(xiv) until cycle = Maximum Cycle Number (MCN)

5.2. Application of ABC

The abilities of ABC algorithm have been previously discussed by a few of researchers. In the research of Rao and Pawar (2010b), the researchers employed seven steps to optimize the process parameters of multi-pass milling operation. The steps in ABC optimization of process parameters are (i) parameter selection, (ii) calculate the nectar amount of each food source, (iii) determine the probabilities by using the nectar amount, (iv) calculate the number of onlookers bees, which will be sent to food sources, (v) calculate the fitness value of each onlooker bee, (vi) valuate the best solution and (vii) update the scout bee. Various WEDM parameters such as pulse-on time, pulse-off time, peak current, and servo feed setting were optimized. A mathematical model was developed using RSM and the machining performance measured is machining speed and surface roughness. ABC was employed to find the optimal combination of process parameters with an objective of achieving maximum machining speed for a desired value of surface finish.

Optimization of process parameters of advance machining process known as ultrasonic machining (USM) was considered in the study of Rao, Pawar, and Davim (2010a). The machining performance is MRR. Five process parameters, amplitude of ultrasonic vibration, frequency of ultrasonic vibration, mean diameter of abrasive particles, volumetric concentration of abrasive particles, and static feed force, and three evolutionary optimization techniques, ABC, PSO and harmony search (HS), were considered in the study. The results of the presented algorithms are compared with the previously published results obtained by using GA. In the study of (Rao & Pawar, 2010a), process parameters of grinding process were considered for optimization such as wheel speed, workpiece speed, depth of dressing, and lead of dressing. The machining performances are production cost, production rate, and surface finish subjected to the constraints of thermal damage, wheel wear, and machine tool stiffness. ABC, PSO and HS optimization techniques were used in this study. The results of the algorithms were compared with the previously published results obtained by using other optimization techniques.

Table 3
Summary of recent PSO techniques in optimizing machining process parameters.

No.	Author/year	Process parameters	Machining process	Machining performance	Remarks
1.	Bharathi and Baskar (2011)	Cutting speed, feed, depth of cut	Turning	Machining time, surface roughness	PSO is capable of selecting appropriate machining parameters for turning operation.
2.	Farahnakian, Razfar, Moghri, and Asadnia (2011)	Cutting speed, feed, depth of cut	End milling	Cutting forces and surface roughness	A very good training capacity of the proposed PSONN algorithm
3.	Yang et al. (2011a)	Number of passes, depth of cut in each pass, speed, and feed	Multi-face milling	Production cost	The proposed schemes may be a promising tool for the optimization of machining process parameters
4.	Yang et al. (2011b)	Number of passes, depth of cut in each pass, speed, and feed	Multi-pass face milling	Production cost	The F-MOPSO does not have any difficulty in achieving well-spread Pareto optimal solutions with good convergence to true Pareto optimal front for multi-objective optimization problems
5.	Costa et al. (2011)	Cutting speed, feed, depth of cut	Multi-pass turning	Production cost	The performance of the proposed technique was compared with other existing methods
6.	Ganesan et al. (2011)	Depth of cut, cutting speed and feed	Multi-pass turning	Production time	PSO produce better results than GA
7.	Razfar et al. (2010)	Cutting speed, feed, depth of cut, engagement	Face milling	Surface roughness	A good agreement is observed between the values predicted by the PSONNOS algorithm and experimental measurements
8.	Zheng and Ponnambalam (2010)	Feed rate, cutting speed, depth of cut	Multi-pass turning	Production cost	PSO performs better than GA and SA
9.	Rao et al. (2010a)	Amplitude of ultrasonic vibration, frequency of ultrasonic vibration, mean diameter of abrasive particles, volumetric concentration of abrasive particles, and static feed force	USM	MRR	The results of the presented algorithms are compared with the previously published results obtained by using GA
10.	Rao and Pawar (2010b)	Number of passes, depth of cut, cutting speed and feed	Multi-pass milling	Production time	The results are compared with the previously published results obtained by using other optimization techniques
11.	Bharathi and Baskar (2010)	Number of passes, cutting speed, feed, and depth of cut	Single pass turning multi-pass turning, and surface grinding	Production cost, MRR	From the results PSO give the best results compared to GA and SA in the three turning operation
12.	Xi and Liao (2009)	Feed rate, cutting speed	Turning	Machining time, machining accuracy and machining cost	The optimized cutting parameters values are better meet the user's optimization goals
13.	Escamilla et al. (2009)	Speed, feed and depth of cut	End milling	Surface roughness	PSO optimization it can be successfully applied to multi-objective optimization of titanium's machining process
14.	Ciurana et al. (2009)	Laser fluence, position of focal plane, laser spot size. translation distance between subsequent laser pulses	Pulsed laser micro machining	Surface roughness, volume error	The proposed models and swarm optimization approach are suitable to identify optimum process settings
15.	Prakasvudhisarn et al. (2009)	Speed, feed and depth of cut	CNC end millling	Surface roughness	Both techniques can achieve the desired surface roughness and also maximize productivity simultaneously.
16.	Srinivas et al. (2009)	Feed rate, cutting speed, depth of cut	Multi-pass turning	Production cost, machining time	The best solution in each generation is obtained by comparing the unit production cost and the total non-dimensional constraint violation among all of the particles
17.	Rao et al. (2008)	Tool feed rate, electrolyte flow velocity, and applied voltage	ECM	Dimension accuracy, tool life, metal removal rate	The proposed algorithm are compared with the previously published results obtained by using other optimization techniques
18.	Li et al. (2008)	Spindle speed, feed rate	Milling	Cutting force, tool-life, surface rougness and cutting power	PSO in optimizing process parameters can converge quickly to a consistent combination of spindle speed and feed rate
19.	Duran et al. (2008)	Cutting speed, power, feed speed, depth of cut	Various	Tool geometry	The selection of the appropriate cutting tool geometry is possible in real world environments
20.	Chen and Li (2008)	Depth of cut, feed rate, grit size	Grinding	MRR	The proposed algorithm is an effective method for grinding process optimization problem
21.	Zhao et al. (2008)	Spindle speed and feed rate	Milling	Cutting forces	The machining process with constant cutting force can be achieved via process parameters optimization based on virtual machining
22.	Liu and Huang (2008)	Feed and cutting	N/A	Cost performance	PSO is relevant for solving complicated nonlinear problem
23.	Tang et al. (2008)	Spindle speed, feed, and depth of cut	Single and multipass turning	Machining time	The proposed algorithm is superior to the latter not only in terms of computational time but also in terms of performance
24.	Zŭperl, Cŭs, and	Cutting speeds and feed rates	Milling	Cutting forces	Compared with GA and SA the proposed

(*continued on next page*)

Table 3 (*continued*)

No.	Author/year	Process parameters	Machining process	Machining performance	Remarks
	Gecevska (2007)				algorithm can improve the quality of the solution while speeding up the convergence process
25.	Huang et al. (2007)	Spindle, Feed rate, width	End milling	Tool wear	The MPSO-trained WNN has a superior performance than BP-NN, conventional WNN, and GA-based WNN

ABC was employed by Samanta and Chakraborty (2011) to find the optimal combinations process parameters of modern machining such as ECM, electrochemical discharge machining (ECDM) and electrochemical micromachining (ECMM) processes. The results obtained while applying the ABC algorithm for parametric optimization of these three NTM processes are compared with those derived by the past researchers, which prove the applicability and suitability of the ABC algorithm in enhancing the performance measures of the considered NTM processes. The machining performance measure include metal removal rate, radial overcut and heat affected zone. The review of the latest researches in optimizing process parameters of traditional and modern machining using ABC techniques was summarized in Table 4.

6. Ant colony optimization

ACO algorithm was inspired by the behaviour of the ants in searching of their food sources. The original concept of ant system is introduced by Marco Dorigo in 1992. In ACO, the ant search for the foods and evaluates the food sources and brings it back to the

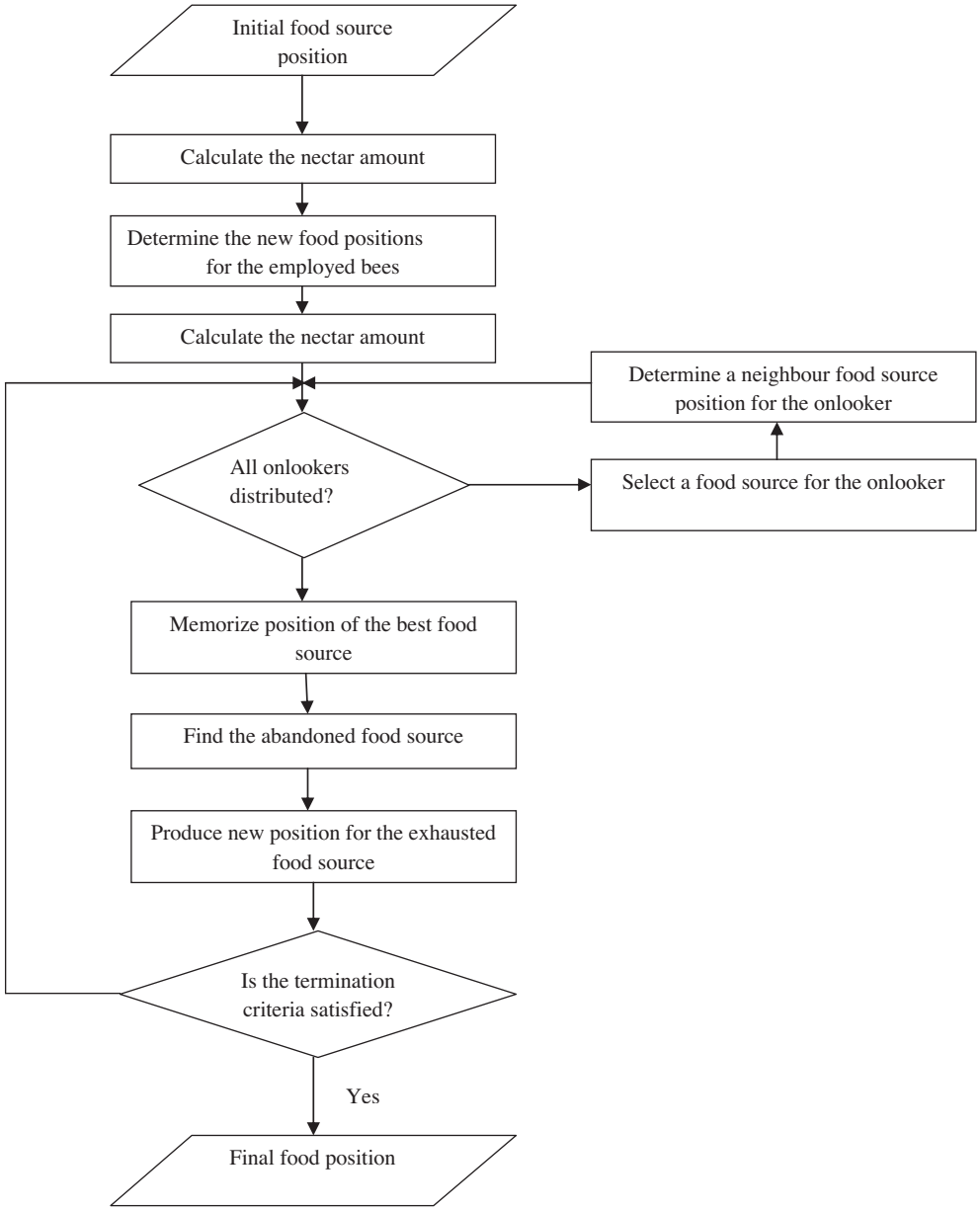

Fig. 6. Flow of ABC optimization (Karaboga, 2009).

Table 4
Summary of recent ABC techniques in optimizing machining process parameters.

No.	Author/year	Process parameters	Machining process	Machining performance	Remarks
1.	Samanta and Chakraborty (2011)	Electrolyte concentration, electrolyte flow rate, applied voltage Inter-electrode gap	ECM, ECDM, EMM	Metal removal rate, radial overcut, heat affected zone	It is shown that the ABC algorithm can also efficiently solve the multi- objective optimization problems of the modern machining processes
2.	Rao et al. (2010a)	Amplitude of ultrasonic vibration, frequency of ultrasonic vibration, mean diameter of abrasive particles, volumetric concentration of abrasive particles, and static feed force	USM	Metal removal rate	The results of the presented algorithms are compared with the previously published results obtained by using GA
3.	Rao and Pawar (2010a)	Wheel speed, workpiece speed, depth of dressing, and lead of dressing	Grinding	Production cost, production rate, and surface roughness	The results of the algorithms were compared with the previously published results obtained by using other optimization techniques
4.	Rao and Pawar (2010b)	Number of passes, depth of cut, cutting speed and feed	Multi-pass milling	Production time	The results are compared with the previously published results obtained by using other optimization techniques
5.	Rao and Pawar (2009)	Pulse-on time, pulse-off time, peak current, and servo feed setting	WEDM	Machining speed and surface roughness	ABC was employed to to find the optimal combination of process parameters with an objective of achieving maximum machining speed for a desired value of surface finish

nest. The ant then leaves a substance named pheromones as their move back to the nest. The quantity of pheromone deposited, which may depend on the quantity and quality of the food, will guide other ants to the food source (Dorigo & Blum, 2005). The other ants tend to follow the paths where pheromone concentration is higher.

6.1. ACO methodology

There are three main function in the ACO technique, which are (i) Autosolution construct( ), (ii) PheromoneUpdate( ) and (iii) DaemonAction( ). ACO has been used to solve many combinatorial optimization problems and other problems which have been demonstrated by Martens et al. (2007), Baucer, Bullnheimer, Hartl, and Strauss (2000), Stützle (1997) and Hu, Zhang, Xiao, and Li (2008). According to Venayagamoorthy and Harley (2007), ACO has an advantage compared to GA and SA when the graph may change dynamically, since the ant colony algorithm can be run continuously and adapt to changes in real time. The computational flowchart of ACO is depicted in Fig. 7.

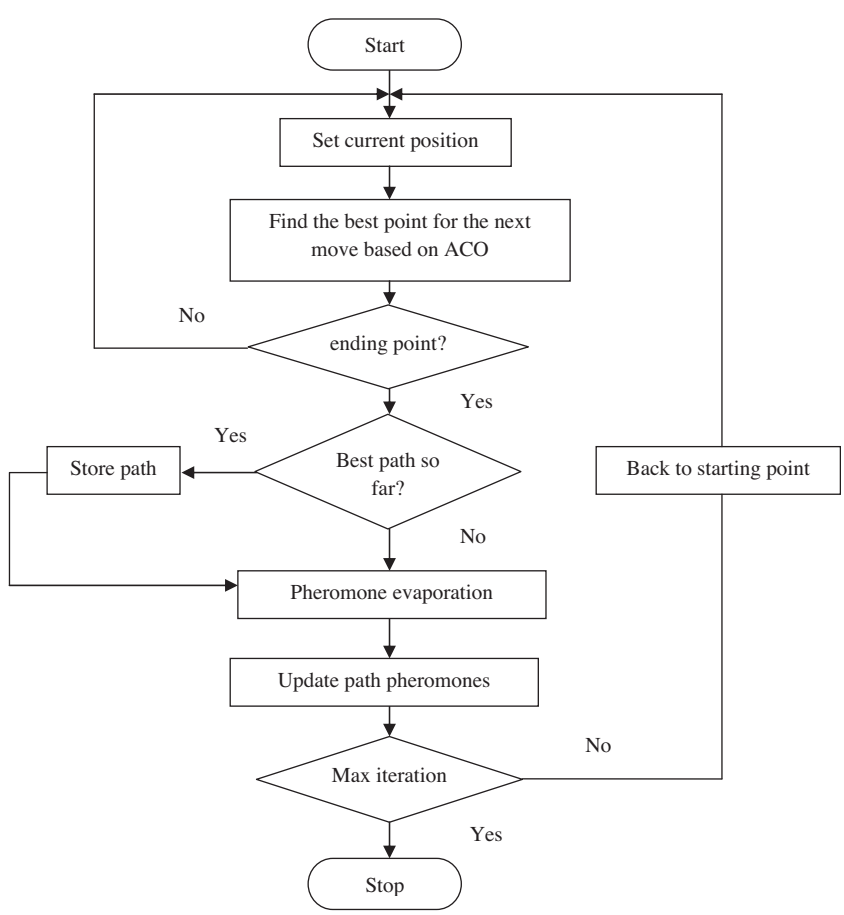

Fig. 7. Computational chart of ACO (Brand, Masuda, Wehner, & Yu, 2010).

Table 5
Summary of recent ACO techniques in optimizing machining process parameters.

No	Author/year	Process parameters	Machining process	Machining performance	Remarks
1.	Kadirgama et al. (2010)	Speed, feed rate, axial depth and radial depth	End milling	Surface roughness	The feed rate was the foremost factors affecting the surface roughness
2.	Cũs, Balic, and Zũperl (2009)	Speed, feed rate and depth of cut	Turning	Production cost Production rate	The proposed ANFIS-ACO approach outperforms GA and SA with 16.02% and 23.08% improvement, respectively
3.	Wu and Yao (2008)	Speed, feed rate and depth of cut	Multi-pass turning	Production cost	The researchers suggested the proposed technique for rapid cutting parameter selection

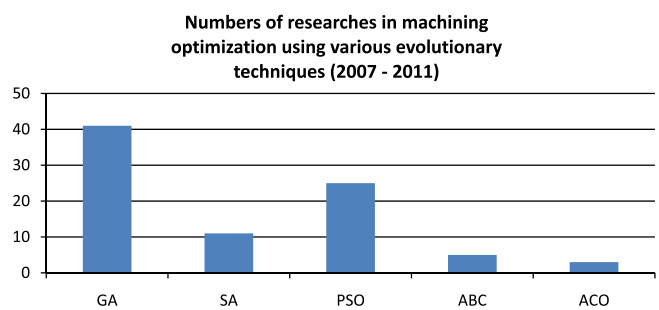

Fig. 8. Numbers of researches in machining optimization using various evolutionary techniques (2007–2011).

In Kadirgama, Noor, and Alla (2010), the researchers applied the combinatorial optimization problems (COP) model to find optimal surface roughness of end milling machining which consists of:

(i) a search space S defined over a finite set of discrete decision variables;
(ii) a set Ω of constraints among the variables;
(iii) an objective function $f : S \rightarrow \mathcal{R}_0^+$ to be minimized.

The main features are the updated pheromone values by all the ants that have completed the trip. The pheromone update for τij (edge joining cities i and j), is calculated by Eq. (8):

$$Tij \leftarrow (1-\rho)\cdot Tij + \sum_{k=1}^{m} \Delta T_{ij}^{k} \tag{8}$$

where ρ is the evaporation rate, m is the number of ants, and $\Delta Tij\ k$ is the quantity of pheromone per unit length laid on edge (i,j) by the kth ant (Dorigo, Maniezzo, & Colorni, 1991) as shown in Eq. (9):

$$\Delta T_{ij}^{k} = \begin{cases} \frac{Q}{L_k} & \text{if any } k \text{ used edge } (i,j) \text{ in its tour} \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where Q is a constant and L_k is the tour length of the kth ant.

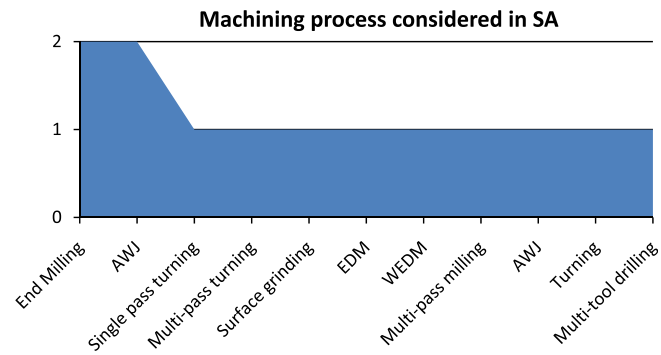

Fig. 10. Machining process considered in SA.

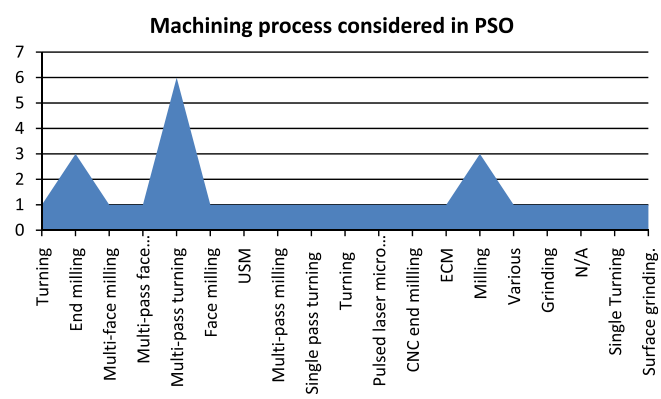

Fig. 11. Machining process considered in PSO.

6.2. Application of ACO

ACO technique has been considered by Cũs, Balic, and Zũperl (2009) to optimize the process parameters of turning process. In this study, the modelled machining performances were production

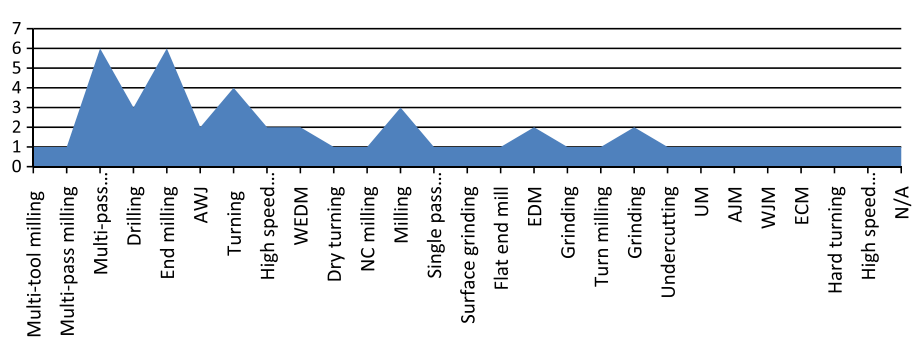

Fig. 9. Machining process considered in GA.

cost and to maximize production rate (which represented by manufacturing time and cutting quality). The process parameters include cutting speed, feed rate and depth of cut. The proposed approach uses adaptive neuro-fuzzy inference system (ANFIS) system and an ACO algorithm to obtain the optimal objective value. From the experiment, it was found out that PSO outperforms all other algorithms compared to SA, GA, and ANFIS-ACO with the minimal production cost = $12.235. However, the proposed ANFIS-ACO approach outperforms GA and SA with 16.02% and 23.08% improvement, respectively. In the research of Wu and Yao (2008), the researchers presented a cutting optimization model for multi-pass turning operation. A meta-heuristic technique, modified continuous ACO (MCACO), has been proposed to find the optimal machining parameters such as cutting speed, feed rate and depth of cut in order to minimize the unit production cost. From the experiment results, it was found that the proposed technique improved the unit production cost compared to other such as float encoded GA (FEGA), SA, ant's colony optimization technique (ACO), hill climbing (HC) and Newton's method (NM). By using the proposed approach, the researchers also found out that the best production cost was $2.203707.

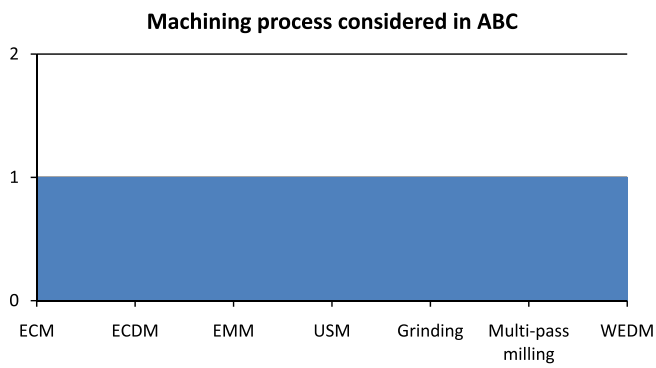

Fig. 12. Machining process considered in ABC.

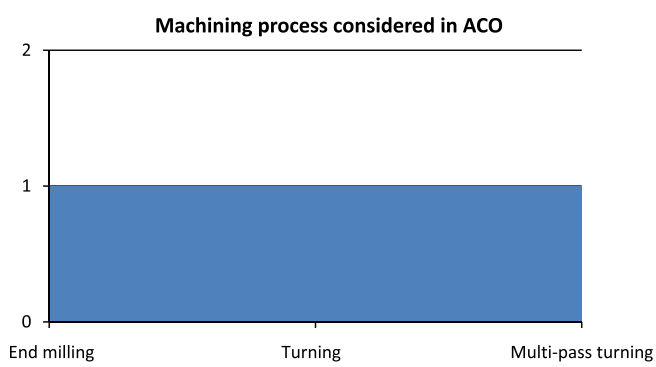

Fig. 13. Machining process considered in ACO.

The technique of RSM and ACO was employed by (Kadirgama et al., 2010) to find the optimal surface roughness in milling mould aluminium alloys (AA6061-T6). The process parameters chosen in this study were cutting speed, feed rate, axial depth and radial depth. From the experiments, the researchers found out that the feed rate was the foremost factors affecting the surface roughness. The errors of surface roughness are 4.65%. The optimal combination of process parameters htat were obtained for minimizing surface roughness are cutting speed = 100 m/min; feed rate = 0.2 mm/rev, axial depth = 0.1 mm and radial depth = 5 mm. The latest researches in optimizing process parameters of traditional and modern machining using SA techniques is shown in Table 5.

7. Discussions and conclusions

From review, we found that GA optimization evolutionary technique is widely used in optimizing machining process parameters followed by PSO, SA, ABC and ACO as depicted in Fig. 8. The researches in machining optimization using latest optimization techniques such ABC only started in 2009 and mostly focused on optimizing process parameters of modern machining such as WEDM, ECM and as illustrated in Fig. 12. For ACO, we discovered there is not as much of researches in machining optimization using this technique from 2007 to 2011. The use of GA and PSO, the most machining operation employed was Multipass-turning as depicted in Figs. 9 and 11, respectively. Fig. 10 confirmed that the most machining process considered in SA technique were end milling and AWJ. For ACO technique, there were three machining processes considered; end milling, turning and multipass turning as depicted in Fig. 13. In Fig. 14, the most machining performances considered by the researchers are surface roughness followed by machining/production costs and MRR. The application of

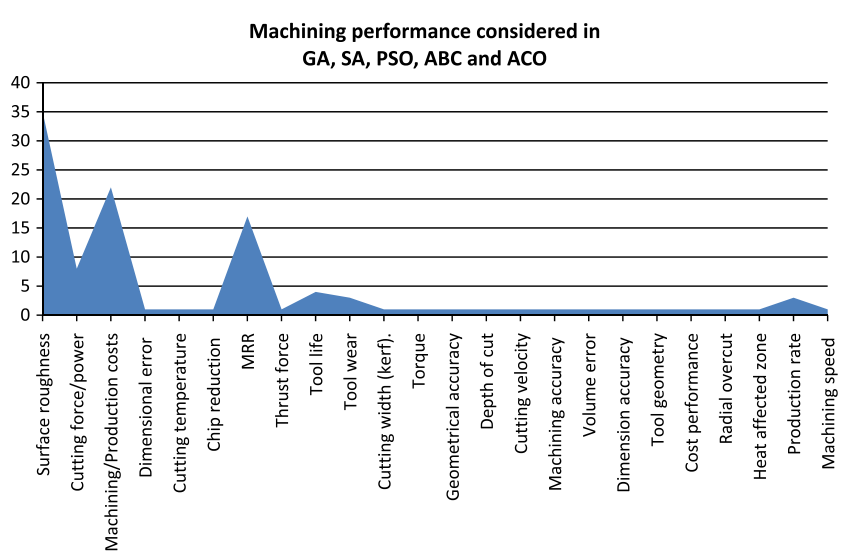

Fig. 14. Machining performance considered in GA, SA, PSO, ABC and ACO.

evolutionary techniques in optimizing machining process parameters positively gives good results as proven from the literature.

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